

# The Ten Earliest Preserved Letters

*To and from Tycho Brahe, 1568–1576 · English translations*

## **1. Tycho Brahe to Johannes of Aalborg [Hans Aalborg], dated 14 January 1568.**

*1568-01-14 · Rostock [Old Style]*

Dearest Master Johannes, know that I have reached Rostock here safe and sound, on the very Kalends of January [1 January], and that in the meantime I was taken in as a guest at the house of Master Levinus. Today, however — God assenting and granting it to prosper — I have resolved to move into lodgings of my own, in the Jurists' College, commodious enough, and well suited to celestial observations and to my own wishes. There, if it shall so seem good to God, I have resolved to pass this winter; what is to be hereafter, the day will bring, and I leave those things for the Lord to look to, mindful of that wise old verse:

*Tend thou to the present; commit the future to the Lord.*

But it falls to you, my Johannes, to bury in deep silence within yourself those reasons for my departure that I confided to you — lest anyone suspect, or come to understand, that I am complaining of anyone about anything, or that there was something in my homeland that drove me to leave. For it is of the very greatest importance that no one hear that I complain of any matter — just as, in truth, I cannot complain of much. For in my homeland I was well received, beyond my deserts, by all my kinsmen and friends; the one thing wanting was that my pursuits did not please everyone — and that deserves its own excuse. There is many an informer everywhere.

Tycho Brahe

## 2. Tycho Brahe to Anders Sørensen Vedel. Greeting.

1571-05-18 · Helsingborg [Old Style]

I have received and read your letter, full of the sweetest consolations, which refreshed me not a little when I was stricken with great grief at my dearest father's departure out of this life; and although I could wish it had been possible for it to be written on some other matter, yet, because that [grief] had befallen me beyond all expectation, [your words] were exceedingly welcome to me. And although there are many consolations — theological, sought from the divine oracles of Holy Scripture, and philosophical, drawn from the common lot of mankind and from the instability of sublunary things and their perpetual vicissitude — with which I lift up my afflicted and grieving mind, yet I cannot but take the greatest consolation from this: that my father so gently and piously passed out of this pitiable wretchedness of mortals into the heavenly and everlasting homeland, where, on Paul's witness, we have a firm dwelling and a city, while here we are all exiles. For it can scarcely be written sufficiently with how great confidence and with how great lightness he met his day [of death]. For when on that same day (which was 9 May), before noon, he had received the body and blood of the Son of God in Bread and Wine, according to the testament and institution of Christ our Saviour, and had fortified himself with great confidence by the sayings of Holy Scripture against the terrors of death and the pains of conscience, he then slept until midday or a little beyond; then at last, waking, he gave signs of the death-struggle now at hand, whereupon we all ran up in dismay, and the ministers of the divine word were called to console him; and then he began as it were to fall asleep. At last, his eyes opened, when he saw us sad and weeping standing about the bed, he asked whether any signs of death were to be seen in him — for he said that he did not feel pains so great but that he had endured far greater before; some of us answered that he was very weak, and that we feared he might not recover. He replied: "Then let me first speak with my wife." When she had been brought, half-dead [with grief], and he had given her thanks for having lived with him so well and honourably in this life, he bade her farewell, and commended her to the care of us children and of his other friends; afterward, his eyes again a little closed, he listened to the preacher's consolations, and when the man spoke many things at length, he himself at last, opening his eyes, uttered these words: "One thing more I wish to say. Does not the Lord himself say: *I desire not the death of the sinner, but rather that he be converted and live?*" With this ending he himself closed his eyes and so gently breathed his last that the bystanders

doubted whether he had died — for he used to sleep much more heavily than he then was dying, and in the very death-agony he did not once move his body, nor feet, nor arms or hands or head from their place, but so peacefully and gently fell asleep that the preachers themselves bore witness that they had never seen a more peaceful end of any man. This happy and good death of my father, then, lifts me greatly in my griefs, and for that reason too I have written of it more copiously, because the very recollection as it were refreshes me; and I do not doubt that you will read these things not unwillingly and without weariness, for I judge that we have here lived well only then, when we die well. Now he laboured most grievously, as you know, under a disease which they reckoned to be anasarca [general dropsy], but I [reckoned to be] dropsy of the lung, for his breathing was difficult, his cough most severe, and there was a very great swelling and inflammation of the foot, while in those [other parts], in the arms, he was emaciated and was, so to speak, being consumed in them, to put it in common terms. Last of all, upon all these evils there supervened a most acute tertian fever, in whose third paroxysm he was extinguished on the 9th of this month, as I said, between one and two o'clock in the afternoon, at the very point of the full moon. May the eternal God be merciful to his soul (as he is), and grant to us all a happy and salutary departure out of this calamitous life, for the sake of the Son, the mediator. Amen. Dearest Master Anders, you write that I should send you a certain water suited for curing fevers; although I have none, nor do I know that you saw such a thing in my keeping, yet I shall not fail to share with you a remedy no less good than sure and easy for the removal of fevers, which I could scarcely obtain from an excellent friend, a physician, who by this method has removed all fevers except the quartan, though I had taught him four secrets of mine in its stead. It is as follows: Take equal parts of these two herbs, centaury and blessed thistle, and boil [them] in the distilled water of one of these, or in water of chicory, or even in common water; then to this decoction add a little powder from the lower jaw of the wolf-fish [Lupus piscis], and give it before the paroxysm; and do not reveal this medicine to everyone, for though it is very slight and easy, it is nonetheless in its effect sure and wonderful. Many too are cured by carrying at the neck a large live spider enclosed in a nutshell, until it dies — and that by a magnetic [sympathetic] remedy; this also you can use. I know, moreover, other secret and sure remedies for removing fevers of every kind, but I would rather share these in person, if there be need, than write them in a letter. Fare you well. I send back the songs, for now it is rather a time for weeping than for singing. Greet Master Johan Thomesen in my name.

Helsingborg, 18 May, year [15]71.

Tycho Brahe

**3. To Master Tycho Brahe, son of Otte, most richly adorned with true nobility and with the knowledge of the divine science of mathematics, his lord and friend to be honoured — greeting.**

*1571-11-13 · Augsburg [Old Style]*

I have received three letters from Your Excellency, O most fortunate Tycho [ε■τυχ■στατε Τυχ■ν], most welcome and most delightful to me — both because they reported that affairs at home and your own were prospering, and because they bore witness to your singular regard for me, in which I take the greatest pleasure. For although nature [bids us] bear a father's death with grief, yet Philosophy bids us bear it with moderation — and the more so since now, become your own master, with the Cyclopean Court avoided as if it were Scylla and Charybdis, you have the fullest freedom to live for yourself and the Muses, or rather for Christ: and to a sensible man what could be more to be wished than that? In your deliberation about a threefold journey it is not for me to prescribe anything to your prudence, and perhaps some counsels are fitter for the young, others for the old. Yet were I in your place — you who excel in both wealth and rank, and are in the prime of life — I would think of marriage, and of safeguarding your estate and carrying it on to posterity, unless perhaps for particular reasons this advice displeases you. Let Philosophy and the divine science of mathematics stand to you in a wife's place, until it shall please you to add a wife to them. At home, to be sure, those Hippocentaurs of yours plague you with many affairs foreign to your disposition; and if you cannot endure them, and perceive that they bar your path of virtue, you will act not unwisely if by a change of place you escape such monstrous troubles. And if then your mind is set on roaming quite outside your homeland, you will pass through the Low Countries and France into Italy, and, having surveyed it, make for Augsburg, and visit us again on your way home. So then once more (unless this climacteric year carries me off) I shall enjoy in person your liberal company, full of all learning, and our friendship — of which I grieved bitterly to be bereft after this departure of yours, since which I have meanwhile suffered grave misfortunes. On 23 November last I came through a great danger. And this year, on 10 April, by I know not what evil mischance, I fell heavily down the steps or stairs by which one climbs to my study, and was hurled right to the very floor, in these new lodgings I have rented from Master Ambrosius, whom you know; and the fall was so great that, my head bloodied and my back all but broken, scarcely rising again, I had to use a surgeon's care for a long time. There were added, too, domestic upheavals

through the treachery of certain men whom I had reckoned my friends; and expenses greater than my income — though I live meanly rather than splendidly — brought me into debt, amid an unheard-of dearth of provisions among us.

My health, moreover, as advancing age brings, keeps sliding into the worse; and above all the stone [lithiasis] and gout of the hand [chiragra] torment me, which I fear will shortly rob me of the power of writing. I do not know what direction in my geniture portends such great evils for me around this time. Perhaps my ascendant is now coming to the body of Mars in ■ Taurus, an earthy sign, as you corrected its place — concerning which you earlier foreboded some evil to me, after you had verified my geniture otherwise than Cyprian Leowitz had done, and fitted my past circumstances more aptly, where too you predicted some things about the future more correctly; and I could wish that now too you would be willing to examine it at your leisure and forewarn me of more. You advise me to lessen my labours and troubles, which you advise both as a friend and wisely; and this I must do, especially now that I am freed from the drudgery of the press. Yet I am not permitted to spend this leisure pleasantly, robbed of the friends in whose company I might delight. For since you left us I have here no one with whom I may converse agreeably and beguile the time in learned discourse — that is, I find no man learned, ingenious, judicious, at once weighty and witty, humane and candid, such as you were to us; all the rest either court honours, or give themselves to gain or to wives — indeed to all these at once, and to others of the kind — and live for these almost alone. And so I must bear my lot patiently, until I am freed from this house of toil. May God, Best and Greatest, keep Your Excellency's life long and flourishing, and increase it in piety, wisdom, and learning — which, as you know, are the true goods, surpassing those the common crowd pursues. Let me know at your leisure what you are now undertaking in astronomy, and whether your mind is to continue these lofty studies; though I do not doubt that you will steadfastly persevere in those things that lead to true and lasting glory. Both the Hainzel brothers, who love you most dearly, send you their warmest and most dutiful greetings.

Farewell.

On the Ides of November [13 November], the year 1571, at Augsburg.

Most devoted to Your Excellency,

Hieronymus Wolf.

#### **4. To the noble Tycho Brahe, his most esteemed lord and friend. Vernal greetings — of hands and of bees [CHEIROS ET MELISSAS].**

*1572-05-07 · Copenhagen [Old Style]*

I have received your letter, dearest Tycho, which I read with great delight. I beg you — while you were pouring it out, what Olympian ray steeped you, that you should fix me too with the same tincture? Was it the sacred sulphur of Venus, or the ethereal Mercury of Bacchus, or that divine salt of Apollo celebrated by the ancients, that breathed upon you? May Urania and Vulcan be no longer favourable to me, if you have not delighted mine exceedingly with that extemporaneous flood. Things long premeditated or digested do not always fall out more happily. Those things must of necessity be extemporaneous and impetuous which, penetrating a solid body in a moment, once penetrated convert it into their own nature — as our fathers, and the thunderbolts of both Meteorologies, have taught us. When I sacrifice to the Sun, whom I worship, I shall as a suppliant pray that he may smite you more often with such heat. Be blessed, and with this sublimity of mind — known to few — grow, until the winged armour-bearer of Jove bears you back into the throne of great Jove, whence you may discern the shapeless ruins of the tottering world. But if this must come to pass, then by all means Gheburah and the evil North Wind are to be shunned: the good South Wind is to be sought, and the life-giving East, where we received life, that gift and talent which has been entrusted to us. This South and East are neither bound by the miseries of places, nor do they decline toward any fixed quarter of the sensible world: they are everywhere — within, without, in all things: in the ancient Iliad, in the treasures of the Universe, in the depths of souls; nay, enter in — this is the Latium and the realm of Saturn sought by many. Not, therefore, toward the sensible South is that civility of the Olympian city to be sought — except in so far as you cannot bear the rigid North, just as perhaps it too will not always bear the splendour and so great a gift, even with yourself (as I have often heard tell) as seer and interpreter. And yet not here alone, where we breathe, does the North Wind nourish the dreaded Barbarism. For even in the farthest East and South the penetrating cold of Boreas now and then scorches souls; and that this side of the world, which mists and an evil Jupiter beset, is also bathed in a pleasant temperateness, there is no doubt — yet known to those alone who gaze upon the light above the Horizon of time and place. Let it therefore be a profanation to shut up the garden of delight within the prisons of times or places. At these cast-off skins the Moon, cleansed of her spots, and the gleaming Beryls, smile. Know thyself, and thou

shalt know all things. To one who looks back into this lens [dioptra] no turbid image shall ever deceive him. So much for silences: now one word. There came to us these past days from Cologne that undaunted Dane of Viborg, of whom mention has often been made in your presence — a Greek, an Arab, a Galenist, a Paracelsian, a sham theologian, a philosopher, a physician, a syrup-monger, a chymist, and more, if you would; yet meanwhile, out of all these, nothing. He, [meaning] to visit Viborg, has long not visited his aged parent; he set out yesterday, [meaning] to return to us shortly. He is angling for a philosophical post, suited to spagyric powders and chymical sweat. He is a marvellous apothecary: he will not displease you — provided he be willing to be honest. If you have a mind to attach some such man to your service, beyond doubt he could be useful to you both at home and abroad. When you see the man yourself, he will speak with you face to face [from his very brow]. Meanwhile, if you yourself do not come hither shortly, write back one word about this very matter. But hear this: the Lovers of the Nymphs relate that you have done violence to the woodland Nymphs — the sacred oaks torn up, and the alders that threaten far and wide, and the trees robbed of life; for these grieve that by their own death (which you so often used to say you had inflicted on them) they are hindered, by you, from being able to rouse life by heat, once other things have been buried and digested. Certainly, that they may be carried over to us by rafts into the South or the West, in the very midst of the North there will be no lack of North Winds.

Farewell; and if you long for a vernal crown, appease the Nymphs.

At Copenhagen, 7 May, the year 1572.

Thine,

Pratensis.



**5. There follows my reply, which consists of a hendecasyllabic poem set down in the manner of prose.**

*1572-05-17 · Herrevad [Old Style]*

**To the Doctor of Medicine Johannes Pratensis, and most especially to his cherished friend. May Vulcan, Urania, and Jove be favorable.**

Our servant, as soon as he had got home, brought the letter sent back by you, dear Pratensis; and when I had read it over, I scarcely think it can be told how great a delight of mind it gave us. For it seemed truly full of a more hidden knowledge of things, and to hold more of the ancient wisdom than it is now the custom (alas!) to teach in the schools. Yet no one, I hope, will envy those [schools] the Barbarism and the strange trifles which [they] have now brought in upon the sweet Muses. For who will call it back? who will lead it back again? He [only] points out the road [to ruin] who, of his own will, prefers to do without the true way. But you, learned Pratensis — heeding them not at all (as is your wont) — climb to higher things, and, surveying with sweet toil the upper approaches alike of Earth and of Heaven, withdraw yourself from the crowd, scorning the things that perish. For greater glory is owed to lofty matters, and little things win no praise. Why do you hold back? A mind born of heaven can brook no hindrance, but, sublime, it pierces to the homes of the Gods, sipping Ambrosia and sweet nectar — which Jupiter himself mixes for the gods, and serves at the same time to the gods (if any there be) [that are] begotten — and so on. But why do I digress and delay with many words? Better now that I send back the answer to your letter, which take in these few [lines].

As soon as that undaunted Dane has come back to me from there, bid him betake himself hither at once; and I desire, myself present, to see him in person, and to learn whether he is suited to our ways, and what he resolves to do — whether he is willing to tend my furnaces, fostering the labors of Vulcan, and to give himself to our studies, and to stay with me for some space of time. Then at last I shall decide what is to come of it.

About the Nymphs, too, you should know, Pratensis, that I have indeed despoiled them well enough of their alders and oaks and their tresses [their leafage]. But that those spoils have not yet been delivered to you, as is fitting, the cause is the Goddess called Alma, and Ceres, mother of the crops: for she, seeing her own labors pressing at the door [the harvest near], forbade the farmers to assist the Nymphs' cause, threatening at the same time that she would not be favorable at that season when the harvest is wont to come, if

they should court the Nymphs on the days antiquity made sacred to Ceres. The poor wretches are forced to obey the Goddess, who grants the soil its fruits each year; they would sooner summon the woodland Nymphs [later] than go without Ceres' gifts. But once the rites are accomplished that they now celebrate for the fruit-bearing Goddess, the farmers must aid the Nymphs; or if it happens to displease anyone to take part in the rites (as may be), that idle fellow, laden with the Nymphs' spoils, shall be sent to you — I will see that he carries what is to be carried. But now there will scarcely be leisure to write more. So, dear Pratensis, farewell. And since you are a physician, it is easy for you to be able to be always well: greet our friends. See that I can get the **cucurbits** [gourd-shaped distilling vessels] which your potter — the tireless Andreas, I hope — is seeing to make for us. Bid him too, I pray, good health. Herrevad, 17 May, in the year 1572.

Tycho Brahe, devoted to you.

**6. The reply to these [letters] has perished; but what follows is [a reply]  
to certain others, which are not at hand.[A]**

*1573-04-16 · Copenhagen [Old Style]*

**7. To the most noble Master Tycho Brahe, son of Otte, the Dane, at the manor of Knudstrup, most outstanding in learning and in the virtues, his lord and friend to be greatly honoured — greeting.**

1573-07-01 · Augsburg [Old Style]

I was unwilling, noble Master Tycho, that my kinsman, as he departed for his homeland, should come to you without a letter from me — both because I knew that letters from friends who esteem you highly would not be unwelcome to you, and because I ardently desire to learn from you what you think about that new star which arose some months ago. It is wonderful how greatly the most learned men, and those most practised in this art, differ in opinion. For some (to say nothing of its substance) wish to call it a comet, others a star on account of its apparent form; others assert that it has a certain intermediate nature. A certain Italian (his name escapes me) most fiercely contends — in a work issued to the public and dedicated to the Venetian Senate — that this heavenly body is not new, but is one of the number of [the stars of] Cassiopeia, [its] light having miraculously been made bright and kindled. Moreover there is great controversy about its distance from the Earth. For some place it in the elementary region — of whose number is our Master Paul Hainzel; others in the aethereal region, with whose opinion Master Peucer, Apian, the Landgrave of Hesse, and others side — who nevertheless also disagree in this, that some place it above the Moon, others above Venus, indeed others do not hesitate to place it in the *primum mobile*, relying on the observations of instruments, on the doctrine of parallax, and on [the doctrine] of triangles. But how can one and the same doctrine, built up out of firm demonstrations, render its practitioners so uncertain that they thus disagree? I beg you again and again to explain these things, and not to find it burdensome — as occasion offers, and for the sake of our common studies — to signify by letter what you determine on this matter. If you do this, it will be as pleasing to me as what is most pleasing; and I shall preserve it forever as a witness of your zeal and affection toward me. Would that we lived together in these times, so that I might have learned the manner of observation from you — in which matter nothing more welcome could have befallen me! But I do not see what I can attain by wishing. Good things, once lost, are for the most part longed for — which I see came about by a certain fate, through no fault of mine, but of the time. The present state of things, then, must be made the best of.

Meanwhile, fare you happily well in the Lord.

Given at Augsburg in Raetia, on the Kalends of July [1 July] 1573.

Yours,

most devoted and most zealous,

Johannes Major.

## 8. To the young man pre-eminent in nobility and learning, Master Tycho Brahe — sends heartiest greeting.

*1574-03-10 · Augsburg [Old Style]*

I cannot tell you, my most noble and most loving Tycho, with what delight you have filled me — not only by the most learned letter lately sent to me, but also by your truly mathematical reflection on that miraculous star by Cassiopeia, which a few weeks ago entirely ceased to be seen, [the star] which I, for my part, have been accustomed to call a comet, in accordance with the opinion received from the school of Aristotle — although, to confess what the matter requires, my own observations, obtained with that great quadrant built here after your prescription, prove that the brightness, whatever it was, was not below the Moon, but in the highest aether, seeing that no parallaxes whatever arise from the observations I diligently set up; and yet I did not dare to say that it had stood above the Moon, lest I affirm anything against the Philosopher, whose authority is paramount, who lays it down beyond doubt that the heaven admits no new bodies. But let experience prevail over authority, especially since you, with your most accurate instruments and observations, have found the same as I — namely, that there was present to this phenomenon [φανομενόν] no parallax at all, just as you have also demonstrated quite evidently in this most learned writing of yours, which indeed far surpasses the conceptions about the same star hitherto published by all others, so that they scarcely deserve to be compared with this one. And I would wish as much as possible that I had obtained it earlier, so that I might have been able to share a description of it with the learned mathematicians hereabouts, while that star yet lasted — especially with Maestlin, who seems to hold the same view as you, although he demonstrates it not so evidently and clearly, but by certain roundabout ways. But I hope that I shall obtain more copies from you, and these — as you seem to promise — rendered fuller; and only then shall I impart them to others. Meanwhile I shall keep this [copy] of mine as a precious thing, and show it to my brother and to our Wolf. As soon, therefore, as you publish more of this kind, do not, I beg, forbear to make me a sharer of them. I have attached to this [letter] my observations made upon this very star, and upon certain others, with the same very great quadrant elaborated here at your prompting, which you will be able to examine, and afterwards to convey to me your judgement upon them. There is no leisure to write more now, on account of the mass of business in this public office which I sustain. So farewell, my excellent Tycho, and do not omit to visit me more often with

your learned letters.

And again, again and again, farewell.

Given at Augsburg, in the year 1574, on the 10th day of March.

Yours,

Paul Hainzel.

## 9. Hieronymus Wolf to Tycho Brahe. He sends heartiest greeting.

1575-03-07 · Augsburg [Old Style]

You have forestalled me, most noble Tycho — robbing me of my own part, which was to give thanks to your singular kindness, in that you do not disdain as a friend an old man who is now nothing but an useless and wretched burden upon the earth. But this, to be sure, is the fruit of true learning: that the loftier you are, the more humbly you bear yourself. For my own part I am most often seized with longing for your most learned conversation, in this solitude that my ill-health affords me. And if from afar I could render any service that would be welcome and agreeable to you, I would do it indeed, both faithfully and gladly. But the treachery of the carrier brought it about that you both lost Master Ambrosius's library [Bibliotheca], and — your money gone — did not receive Schreckenfuchs's armilla. Of these matters you will in any case be informed more fully both by the Magistrate [Maior] and by our Consul — who charged me to learn from Schreckenfuchs whether there were such an armilla, and whether, and at what price, he was willing to sell it; which I shall do at the first opportunity. For there is rarely a chance of sending letters from here to Freiburg. Whence I foresee that a fairish delay will intervene before these things are conveyed to you. But [we do] as we can, since we may not [do] as we would.

What you reply concerning the magnitudes of the stars is welcome; nor am I ignorant that certain men interpret the falling of the stars allegorically, of the lapse of the foremost learned men. But that magnitude of the heavenly bodies, exceeding all the thought of the human mind, often makes me as it were thunderstruck and snatches me out of myself with the utmost amazement — when I compare it with the axioms of our religion, hemmed in within such narrow bounds of both time and place; [a religion] that sets us most vile manikins, foulest and most wretched little animals, not merely on a level with all the adornment of the heavens, but ahead of it by an infinite interval. Indeed, [men] write that nothing seemed more absurd to Julian Caesar than to maintain the destruction of the Sun and the resurrection of earthly little bodies. Master Peter, certainly, in his 2nd Epistle, last chapter, writes expressly: "the heavens that now are, and the earth, are stored up by the word of God, kept for fire against the day of judgement and of the destruction of impious men" [ο■ ν■ν ο■ρανο■ κα■ ■ γ■ θεο■ λ■γ■ τεθησαυρισμ■νοι ε■σ■, πυρ■ τηρο■μενοι ε■ς ■μ■ραν κρ■σεως κα■ ■πωλε■ας τ■ν ■σεβ■ν ■νθρ■πων]. And [things] not unlike are in Ezekiel, Isaiah, and the other prophets —



which do not seem able to be allegorized [■λληγορε■σθαι]. Philo, however, in his little book *On the Incorrutibility of the Cosmos* [περ■ ■φθαρσ■ας κ■σμου], seems to me at least to prove the eternity of the world by reasonings so many and so unconquerable that nothing can be set against them save the authority of the sacred Writings. Yet, in order that we may rightly understand these [authorities], the matter must be looked into again and again. And so I am your advocate both for weighing those passages diligently, and for conferring about them with learned men — not only Philosophers but also Theologians — and above all with Master Hemmingsen, who has joined Philosophy to Theology, and eloquence to both. Certain petty-philosophers too (among whom Thomas Erastus, the Heidelberg physician, triumphs most of all) and petty-theologians assail divinatory Astrology as both opposed to religion and divinely forbidden. To me, however, it seems that nothing assails Christianity more than the infinite height and magnitude of the heavens. I judge that you ought to make mention of these matters in [your work on] Astrology — or rather that those questions ought to be treated with precision. But the things that seem most of all to shake the credit of [astrological] predictions are these, which I leave to you to refute; for both wit and learning fail me in this kind.

For how few there are who observe the altitude of any star! And as for the Ptolemaic conjunctions [■] and oppositions [■] and the *trutina* of Hermes, Antonius Montculmius convicts them of great errors and absurdities. Now some men refer one set of events to one set of significators, others to another; and divers outcomes show a divers Horoscope [ascendant] in my geniture, which is ennobled by very many notable misfortunes and by some successes too. And in the progressions, how great a difference does the latitude, whether applied or omitted, make? But what would you say of Camerarius — a most learned man, and, while he lived, most friendly to me — who subtracts 27 degrees both from the Horoscope and from the motion of each several planet? Thus for me he casts ■ Capricorn as horoscope [ascendant] instead of ■ Aquarius: ■ Saturn is set in ■ Scorpio, ■ Jupiter in ■ Cancer, ■ the Sun [and] ■ Mercury in ■ Leo, ■ Venus in ■ Virgo, ■ the Moon in ■ Libra. A vast difference, surely. It will be for your industry to set forth these and many other matters in such a way that both the more untaught (in whose number I, an old man, enrol my name) may be instructed, and the mouths of shameless slanderers stopped. That philosophic leisure of yours, so contrived that its fruits lie open most widely and far surpass the business of many men [though] arduous in appearance — I congratulate you on it from my heart. And I exhort you (though indeed there is no need

of it) never to suffer yourself to be driven from that most honourable resolve, either by any rouse of courtly splendour or by the mockeries and idle exercises of an ignoble nobility. Go on and prosper in your virtue, Tycho: thus is the way to the stars. As for your exhorting me to equanimity — I acknowledge your goodwill toward me, and I shall obey your most faithful admonitions as far as I can, although by nature little stout. Indeed, with all my manhood I

Strive, but my patience, vanquished, gives way to the pain.

Words lighten the spirit, but pain has no ear;

Stormed by ills, our wisdom totters and slips

While the grievous, the long-drawn sickness racks [me].

Indeed I have read all the consolations both of the Theologians and of the Philosophers; and, while it is well with me, I make light of them [reckon them fair and good]; and the perturbations of the mind, to speak with God's help [σὺ θεὸς ἐπέμψ], I can moderate moderately well — but the pains of the body are relieved not by words, but either by stones and herbs, or by death. For the necessity of nature increases the pain for me as much as for Carneades; nor does Cleanthes, with those magnificent verses of his (which Epictetus cites and Seneca elegantly turns) at all help me as I cry out and wail in my sickness:

Lead me, Father and ruler of the lofty pole,

Whithersoever it has pleased thee. There is no delay in obeying.

Here I am, unflagging. Suppose I will not? I shall follow, groaning,

And suffer in wickedness what I might have suffered as a good man.

The Fates lead the willing; the unwilling they drag.

But enough of this. One thing remains: that I give thanks to your kindness and to the generosity of [your] uncle Steno Bille, in that you have graciously taken in Marcus Hartlieb, my pupil — taught by hardships what it is to despise the admonitions of parents and preceptors. I hope you will lay out your benefactions well [to good effect].

Of news I write nothing, since I know little, and that bad; but what is reported here you will learn from Peter. Northern affairs are surely better known to you; but what is told here of Megalopyrgium [Megalopyrgio], of the Swede, of the Poles, and indeed of Saxony itself, gives little good hope. Amurath [Murad III], the successor of Selim, is said to have had all his younger brothers killed and buried together with their father, [and] to

have granted the Emperor a truce of eight years, meaning to turn the whole weight of war against the Persians. But I fear these are the dreams of the blessed, and that we, off our guard, may be overwhelmed by a most treacherous and most powerful enemy. Farewell, and forgive [me]; and hold it for assured that I shall keep the memory of you inviolate, O most fortunate Tycho [■ ε■τυχ■στατε Τυχ■ν], so long as I am mindful of myself, so long as breath shall rule these limbs.

The Consul sends the armillae, and denies that these trifles of mine will be of use to you, nor that Peter should be burdened with them, since you have instruments far better than these. I, for my part, send you as a gift an Astrolabe (not knowing it to be the same as an armilla) given to me by Schreckenfuchs, together with an astronomical ring once made by Joachim Ringelberg; and if these are not going to be of use to you (which I fear), you will yet allow them to be a keepsake [μνημ■συνον] of your Wolf. Of the rest, Consul Hainzel and Master Maior will write more. I send you the letter given to me by Schreckenfuchs, [I being] an unskilled appraiser of this machine. I have bidden him ask what more he requires.

Again, farewell.

On the Nones of March [7 March], 1575.

Your Hieronymus Wolf.

## 10. To the young man pre-eminent in nobility and learning, Master Tycho Brahe, his singular friend — heartiest greeting.

1575-03-11 · Augsburg [Old Style]

If you are well, it is good, my Tycho; I, for my part, by God's kindness am now in sound health, and have at last been freed from a quartan fever. What you wrote about your astronomical instruments I read gladly; and that matter of the Ptolemaic armillae — the only thing I gathered you now wanted — we have easily obtained from Schreckenfuchs, with the help and letters of our friend Wolf (a trusty messenger having been sent to Schreckenfuchs at Freiburg with a gift in return for those armillae). Peter the goldsmith will carry it from here to you along with his own baggage. And although Copernicus gives a full and elegant account of the construction of this instrument in the second book of his *De revolutionibus*, chapter 14, I nevertheless wished to add to the instrument Johann Regiomontanus's exposition too, on its construction and use — which I had transcribed from Regiomontanus's writings (published at Nuremberg, on the five astronomical instruments: the torquetum, the armillary astrolabe, the great Ptolemaic rule, the cross-staff [*baculus astronomicus*], and the geometric square, with the additions of Johannes Schöner of Karlstadt, in the year 1543) — in case you happened to lack that book, though of that I rather doubt. That worthless courier, who absconded after squandering the money you entrusted to him, has not yet returned; either he will wander as a vagrant and an exile from his homeland, or he will restore what he carried off. Marcus Hartlieb, my wife's nephew by her sister, sent two years ago by his parents to Wittenberg for the sake of his studies, stayed there a year, as it chanced. In the first half-year, or rather more, he conducted himself energetically, as becomes a diligent and free-born youth, and began to apply good effort to his letters. In the latter half-year he was less attentive to his duty; and as soon as his parents learned of this — albeit rather late — they at once recalled him from Wittenberg, that he should set out for Heidelberg to his brother Johannes, Doctor of Both Laws, councillor of the Elector. On the way, having fallen in (that I may write nothing too harsh) with a man who was not good, and having dawdled some weeks on that journey, not without great loss to his studies and his money, he came at last to his brother at Heidelberg with a companion, pretending that he had suffered bodily and ill-health troubles on the road. His companion likewise falsely testified to the same before the brother; and, affirming that he had laid out money for Marcus on the journey, he extorted twenty Joachimsthalers from Master Doctor

Johannes, Marcus's brother. Then Marcus, when he had been admonished and rebuked by his brother for everything he had done up to that point, departed from Heidelberg without his brother's knowledge and against his will, ill at ease in his own conscience. Where he set out for afterward, or where he has detained himself up to now, we do not well know. But Peter the goldsmith, who delivered your letter to me, said that he was at Copenhagen with you on good terms, and applying himself to his studies. Master Johannes Maior also indicated that he had received a letter from Marcus, through the same Peter, in which he wrote much the same thing, and besides begged pardon for his fault and asked that [Maior] would treat with his father about a reconciliation. But the father, being angry, will still be hard to reconcile. If he shall have shown true repentance and come to his senses, you will do me a kindness if, for my sake, you help him — which I should wish to be done without trouble to yourself; but if hereafter he does anything contrary to his duty, and persists in his former negligence, then whatever punishment you exact upon him — even if you chastise him soundly and severely — you will do a welcome thing both to his parents and to us, his kinsmen and relations. His mother, a woman of pious memory, in the past month — namely on 3 February [the third before the Nones of February], at 8 in the evening — fell piously asleep in Christ, having been born into this world 48 years before on the Nones of the same month. This I wished to point out to you: in the construction of the great Ptolemaic armillae (which are capable of [a division into] minutes or of a still finer division), since the distance of the poles (namely of the equinoctial and of the ecliptic) changes with the lapse of time, the little axles, which turn with the zodiacal carrier-ring upon the poles of the world, can be so shifted within the carrier-ring that over a span of 24 minutes (which are, as you know, the differences of the greatest and least obliquity of the zodiac) they may be moved either way at will, in keeping with the change of the Sun's greatest declination, and may nonetheless, anywhere and at any time within such a span, be connected and turned together with the pole of the world along the diameter at a fixed place. Furthermore, this too seems to make for a fuller perfection of this instrument (whereby the places of the stars may be had and taken more accurately and exactly, even to seconds, thirds, fourths, etc. of scruples [of arc]): that that geometric and artful distribution of very many circles, drawn one within another, should be marked and traced out in due manner on the flat sides of the armillae and on their jaws [maxillae] (which can easily be done given so great a breadth of the armillae). But why do I [carry] owls to Athens, in [writing] these things to you? I had nearly omitted to mention this to you: that our quadrant at

Göggingen, on the last Nones of December [5 December], a rather violent wind having risen, slipped and collapsed, and the use of it is now wholly taken from us. Again and again, farewell. The key to the case of the armillae I send here together [with this]. My brother and Andreas Welser return you many greetings, and are wont often to recall your memory with honourable mention of you.

Again, farewell; and do not cease — as you do — to return my love.

Given at Augsburg, on the day of the vernal equinox, that is, the 5th [day before] the Ides of March [11 March], in the year of salvation 1575.

Yours, Paul Hainzel.